

Total no. of Pages: 2

Roll no.

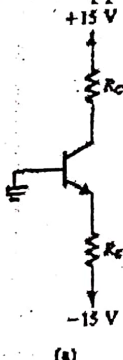
Degree: B.Tech.(ECAM) Semester: 3rd Semester— Course work
MID-SEMESTER EXAMINATION, September, 2025

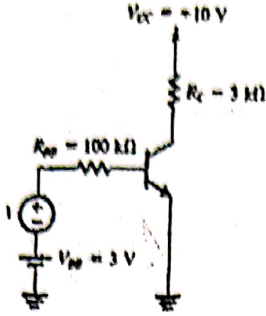
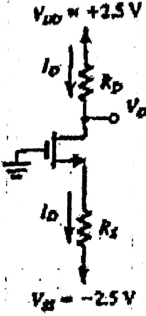
Course Title: Microelectronics
Course Code: EAEPC-07

Duration: 1:30 Hours

Max. Marks: 15

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	Consider an amplifier operating from $\pm 10V$ power supplies. It is fed with a sinusoidal voltage having $1V$ peak and delivers a sinusoidal voltage output of $9V$ peak to a $1-k\Omega$ load. The amplifier draws a current of $9.5 mA$ from each of its two power supplies. The input current of the amplifier is found to be sinusoidal with $0.1 mA$ peak. Find the voltage gain, the current gain, the power gain, the power drawn from the dc supplies, the power dissipated in the amplifier, and the amplifier efficiency.	2	CO1
1b	What is the technique for obtaining linear amplification from an amplifier with a nonlinear transfer characteristic?	1	CO1
2a	Describe the operation of MOSFET as a resistance when we apply a small voltage v_{DS} .	2	CO2
2b	What are the various advantages of MOSFETs over BJTs?	1	CO2
3a	Draw the schematic of Ebers-Moll (EM) model of a npn transistor.	1	CO2
3b	The transistor in the circuit of Fig. (a) has $\beta = 100$ and exhibits a v_{BE} of $0.7 V$ at $i_c = 1 mA$. Design, the circuit so that a current of $2 mA$ flows through the collector and a voltage of $+5 V$ appears at the collector .  (a)	2	CO2
4a	Draw and explain the Hybrid- π and T models of BJT.	1	CO1
4b	Analyze the transistor amplifier shown in Fig. below to determine its voltage gain. Assume $\beta = 100$.	2	CO2

			
5a	How would you describe the inversion layer in MOSFETs?	1	CO1
5b	Find the unknown circuit parameters of Fig. provided below so that the transistor operates at $I_D = 0.4 \text{ mA}$ and $V_D = +0.5 \text{ V}$. The NMOS transistor has $V_t = 0.7 \text{ V}$, $\mu_n C_{ox} = 100 \mu\text{A/V}^2$, $L = 1 \mu\text{m}$, and $W = 32 \mu\text{m}$. Neglect the channel-length modulation effect (i.e., assume that $\lambda = 0$).	2	CO2
			

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